Graphical Plotting in R

Traditional graphics treats each type of plot separately

- Many different functions, each with distinct syntax and arguments
- Hard to learn and remember; hard to read / update code later
- Hard to experiment with different plot types

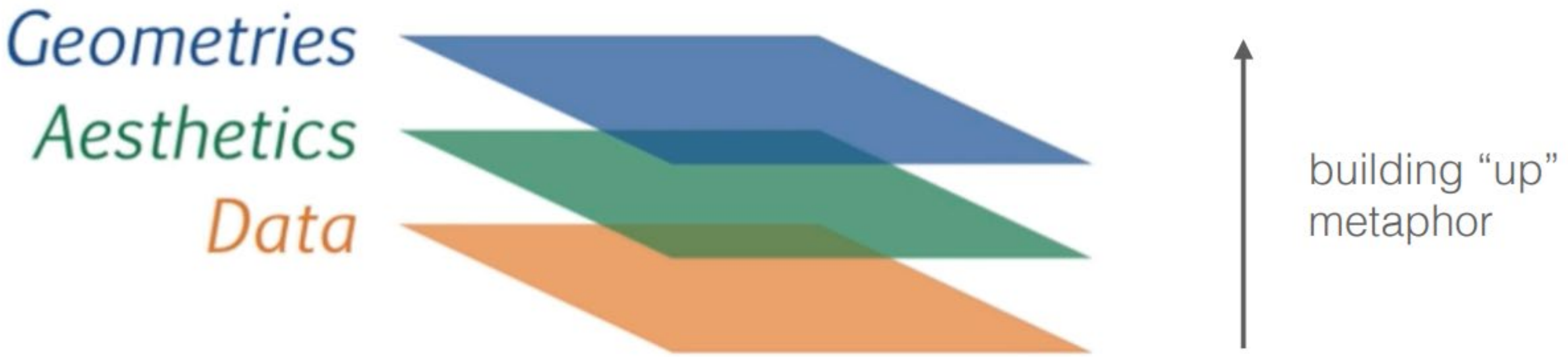
ggplot2 is built around the idea of a **“grammar of graphics”** (Wilkinson 1999)

- Core concept: define plots based on **semantic components**

Important advantages:

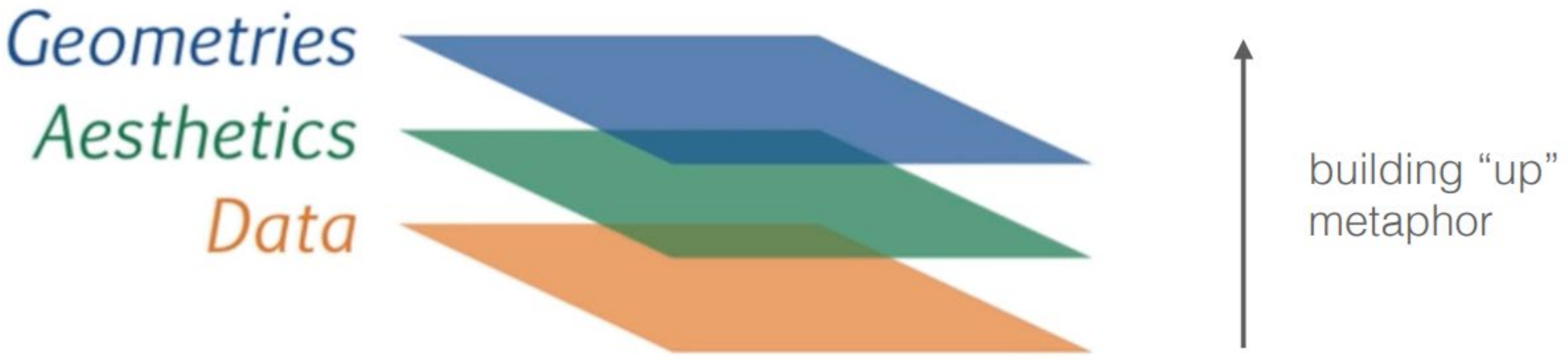
- Uses **consistent syntax** and arguments for all plot types: easier to learn / code
- Allows combination of plot types through **layering**, multiple plots through **faceting**
- Provides customizable **themes**: consistency is easier for audiences / readers to follow

Graphical Plotting in R




```
ggplot(data = frame, mapping = aes(columns)) + geom_X()
```

Graphical Plotting in R

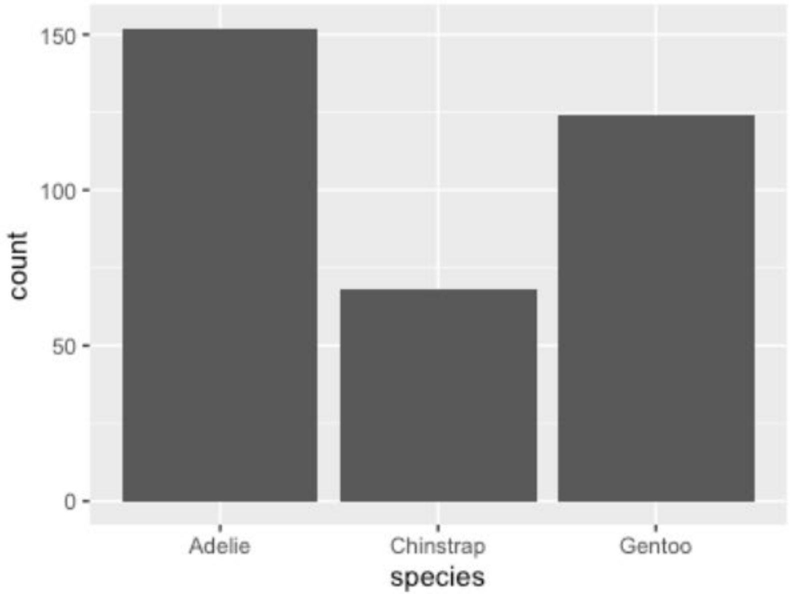


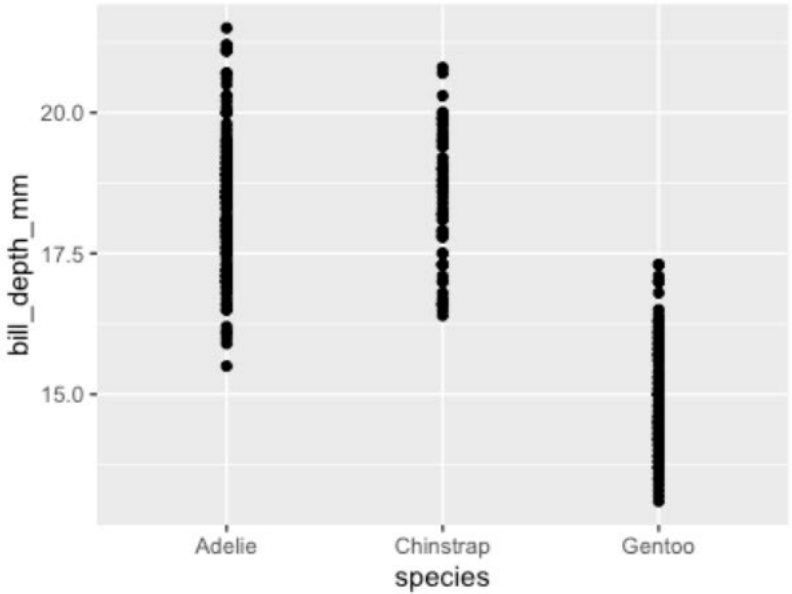

```r
ggplot(data = penguins,
    mapping = aes(x = bill_depth_mm, y = bill_length_mm)) +
    geom_point()
```

aes layer: one vs two layers

```
ggplot(penguins, aes(x = species)) +
    geom_bar()
```

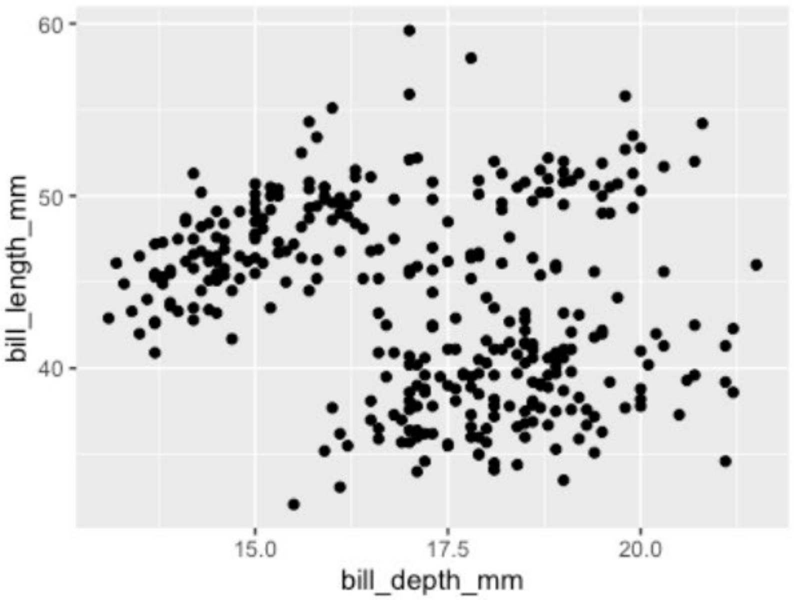
```
ggplot(penguins,
aes(x = species, y = bill_depth)) +
    geom_point()
```

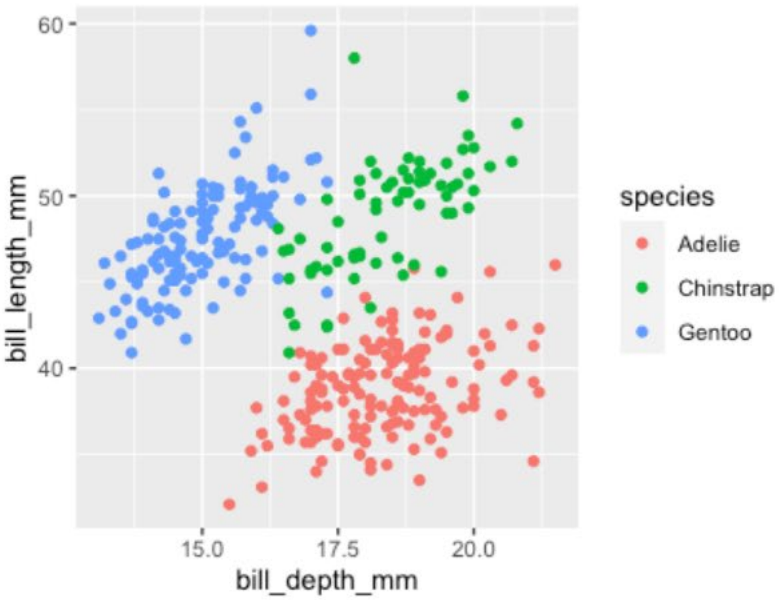




Three variables - color!

```
ggplot(data = penguins,
    aes(x = bill_depth_mm, y = bill_length_mm, color = species)) +
    geom_point()
```



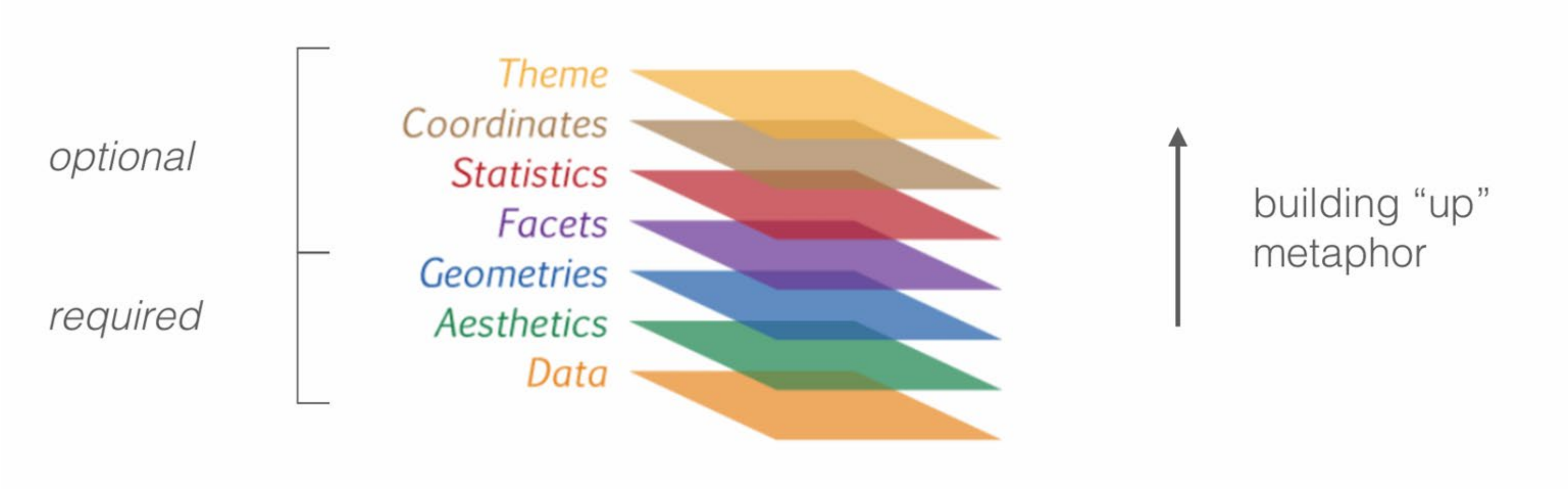


Optional but useful!

Theme: every part of the graph not related directly to data; the “look and feel”

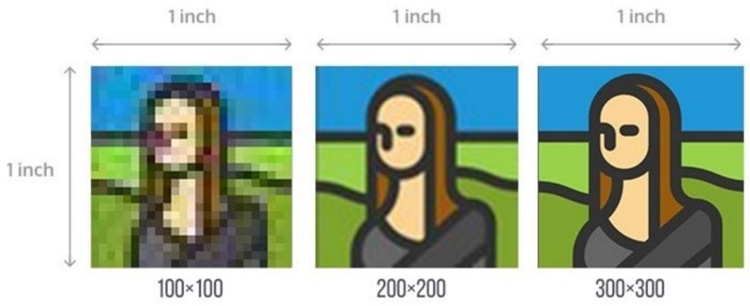
Coordinates: how variables are mapped, scaled, and applied to a particular geometry

Statistics: calculations of derived values; linked to specific geoms Image credit:



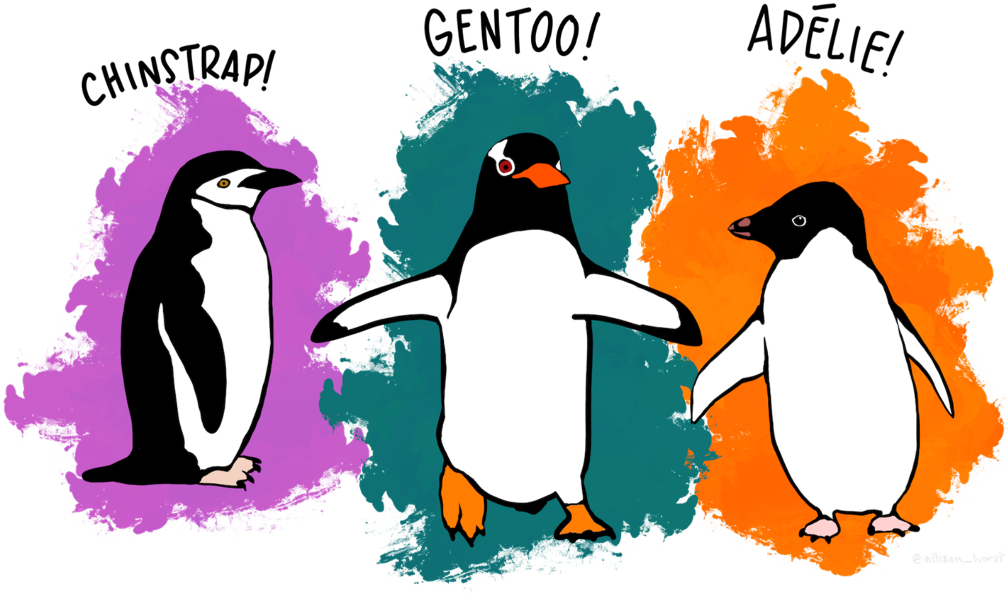
Exporting Basics

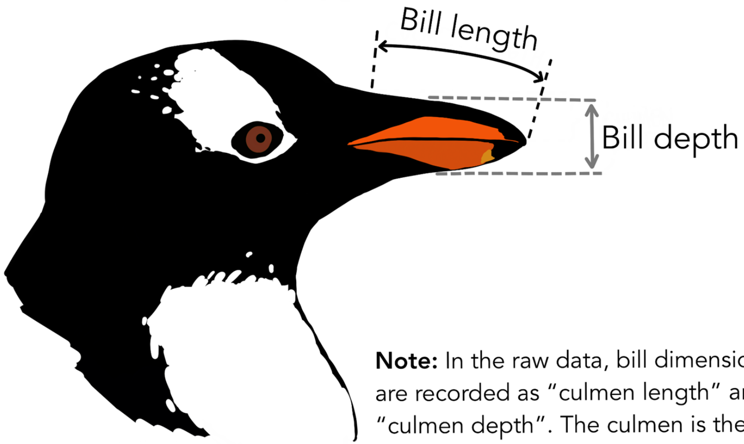
```
png("penguins_bar.png", width = 2000,
height = 1500, res = 300)
print(ggplot(penguins, aes(x = species))
+ geom_bar())
dev.off()
```



File Format	*R Function*	*When to Use*
PNG	png()	High-quality images for slides, web, or reports; supports transparency; lossless but compressed.
JPEG	jpeg()	Smaller file size; good for photos or quick sharing; **not recommended** for plots with text/lines (lossy compression).
TIFF	tiff()	Publication-quality, lossless format; often required by scientific journals. Large file size.
BMP	bmp()	Rarely used; simple, uncompressed raster format.
PDF	pdf()	Vector format; scalable without quality loss; ideal for printing and LaTeX integration.
SVG	svg()	Vector format; editable in Illustrator/Inkscape; good for web and scalable graphics.
EPS	postscript()	Vector format; older standard for publishing; still accepted by some journals.

ggplot with Penguins





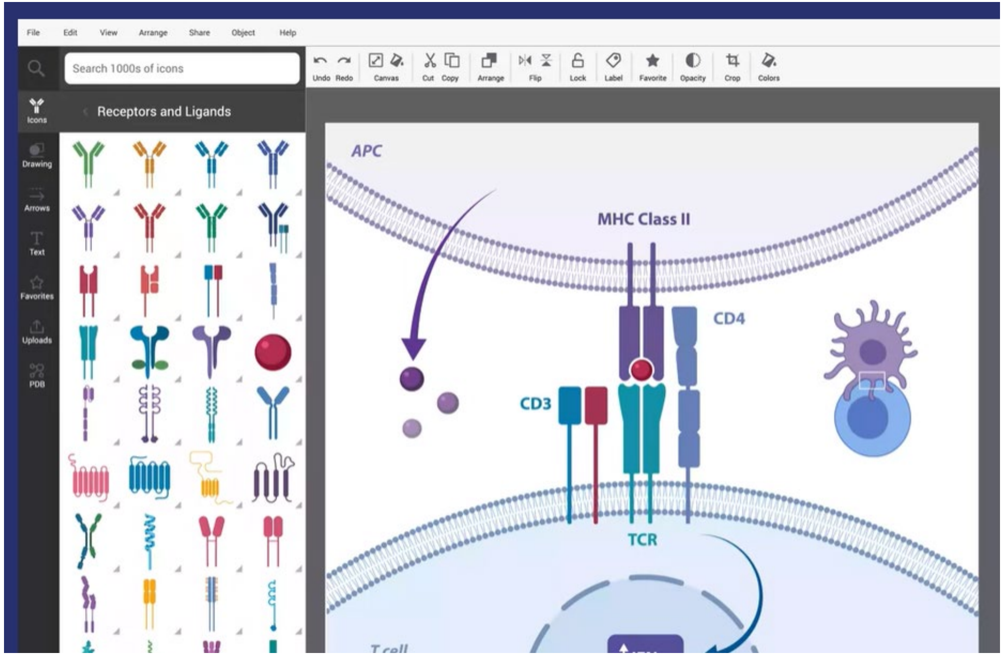
Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.

https://allisonhorst.github.io/palmerpenguins/articles/intro.html

BioRender

https://www.biorender.com/

Duke has departmental subscriptions to BioRender for Chemistry and Evolutionary Anthropology. If you are in an associated lab that has Biology or Biochem affiliations, you may be able to access those subscriptions!



NIH BioArt

https://bioart.niaid.nih.gov/

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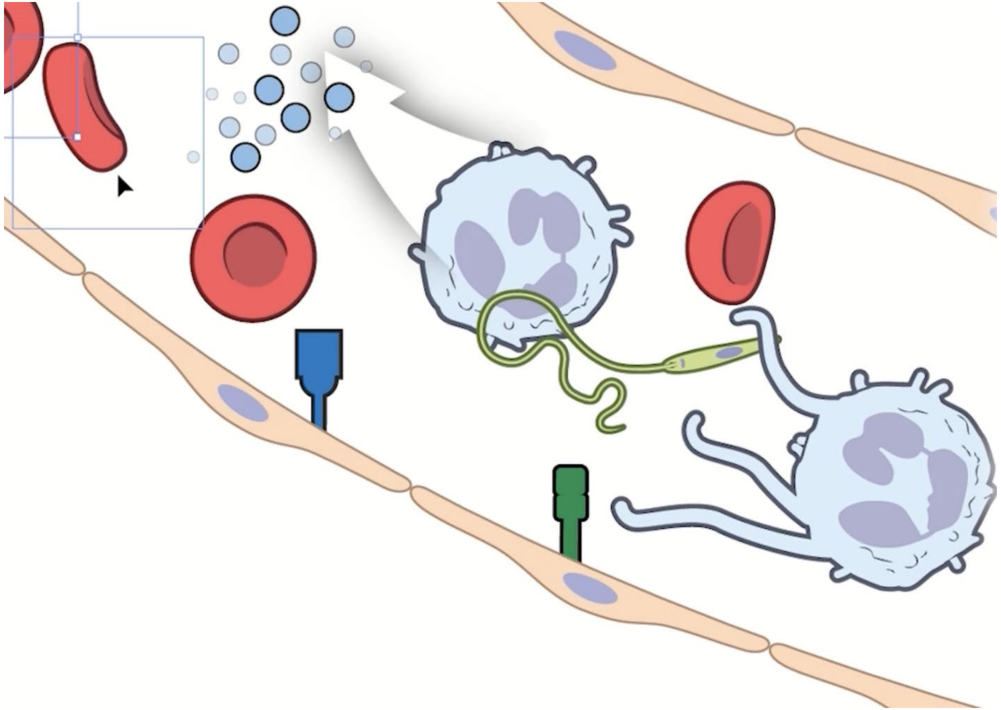

Adobe Illustrator

Adobe Illustrator is a vector graphics design software used to create scalable illustrations, logos, icons, and typography. It’s widely used in graphic design, branding, digital art, and print for its precision and powerful creative tools.

Free Duke access!